

## Response to Referee #1 (Follow-up Comments)

We thank the referee for the additional comments. They have helped us further clarify the scope of the model, the interpretation of the historical decomposition, and the nature of our contribution. Below we respond point by point.

### 1. Technology shocks, capacity utilization, and the interpretation of key episodes

#### Referee comment:

*For the third comment on TFP shock, the authors should mention changes in capacity utilization rate. Please also explain factors (other than production inputs such as capital and labor) contributing to the big changes in output in years 2009 (global financial crisis), 2012 (?), and after 2017 (?) in Ecuador.*

#### Response:

The model does not feature variable capacity utilization, so we do not attribute fluctuations in output to that margin. More broadly, the purpose of the historical decomposition is not to offer a complete structural account of Ecuador’s business cycle, but to clarify the relative contribution of fiscal, oil, and financial shocks conditional on the model structure. As in standard DSGE practice, technology shocks act as residual drivers required to match persistent movements in output growth that stationary shocks cannot generate. We clarify this interpretation in the revised text and footnotes, emphasizing that these shocks should not be interpreted as literal technological changes or precise historical narratives.

In the manuscript, we also reordered the discussion of the historical decomposition so that oil, country risk, and fiscal shocks—our main objects of interest—are discussed more prominently than the technology block. In that discussion, we added the following sentence:

As in any estimated DSGE model, these disturbances should be interpreted as reduced-form residuals capturing persistent forces not explicitly modeled (e.g., institutional changes, sectoral reallocation, or external demand shifts), rather than as literal productivity innovations.

### 2. Dollarization, the trilemma, and external adjustment

#### Referee comment:

*Regarding the fifth comment on monetary policy, the impossible trinity of international finance asserts that monetary policy autonomy, fixed exchange rate flexibility, and free capital account openness cannot be sustained at the same time. Due to the dollarization of Ecuadorian economy, interest parity is automatically applied. In this vein, the manuscript should discuss the other two variables: exchange rate movement and capital account response to shocks.*

**Response:**

Ecuador is fully dollarized, implying a fixed nominal exchange rate and no independent monetary policy. At the same time, capital mobility is not frictionless: domestic borrowing costs differ from the world interest rate through an endogenous sovereign risk premium that depends on aggregate external indebtedness. As a result, uncovered interest parity does not hold mechanically. Capital-account responses are captured through movements in public and private external debt, while exchange-rate adjustment is absent by design.

To make this institutional point more transparent, we added the following footnote in the paper:

Ecuador is officially dollarized, which effectively fixes the nominal exchange rate whereas capital mobility is imperfect in the model, so domestic borrowing costs adjust through the sovereign risk premium or the world interest rate rather than through domestic monetary policy or exchange-rate movements.

### 3. Contributions and novelty

**Referee comment:**

*Comment #10 on novel contributions. The first point is not a contribution because using policy variables decided by the IMF and estimate DSGE responses do not provide any new value added. The episode is already happened in the past, and the IMF macroeconomic framework is already calibrating and estimating such paths of economic and financial variables in the general equilibrium. Please drop this first point from contributions. Also, please cite the existing papers that analyzed how commodity price (oil price) influences the debt dynamics and compare and highlight the different findings from such extant literature. I suspect that the same topic was already heavily explored in the literature.*

**Response:**

We thank the referee for raising this point and would like to clarify the nature of our contribution. While policy institutions routinely produce macroeconomic projections for IMF-supported programs, these projections are typically generated using financial programming or macroframework tools based on reduced-form behavioral equations and accounting identities. Our contribution is to evaluate a concrete, multi-instrument consolidation program within a fully specified structural DSGE framework estimated on pre-program data, which allows us to trace nonlinear interactions among commodity revenue realizations, debt dynamics, and sovereign spreads under perfect foresight.

We would also like to emphasize that we engage directly with the existing literature on commodity prices, fiscal policy, and debt dynamics, which we review in Section 2 of the manuscript. As stated there, our contribution is not to question the importance of commodity prices for fiscal stability—this is well established—but to focus on a different margin. While much of the existing literature studies fiscal policy and commodity revenue volatility in an unconditional or rules-based setting, it does not directly assess how commodity revenue uncertainty interacts with the implementation of a given, pre-announced fiscal consolidation plan.

Our central question is therefore conditional in nature: given a fixed consolidation path for fiscal instruments (as in an IMF-supported program), how sensitive are macro-fiscal outcomes to realizations of commodity revenues? This focus distinguishes our analysis from studies that examine

optimal fiscal rules, generic responses to commodity price shocks, or long-run fiscal sustainability, and speaks directly to the feasibility and risk profile of consolidation programs implemented under revenue uncertainty. We clarify this distinction and adjust the wording of the contribution accordingly in the revised manuscript.

In particular, we replaced the previous wording of the first contribution with the following:

First, it evaluates a multi-instrument IMF-style fiscal consolidation package using a fully specified structural SOE DSGE model estimated on pre-program data. This approach differs from standard financial programming or macroframework projection tools typically used in policy institutions and allows us to trace nonlinear interactions among commodity revenue realizations, external debt dynamics, and sovereign risk premia under perfect foresight.

## Closing remark

We thank the referee again for the additional comments. In response, we have added targeted clarifications to better articulate the scope of the model, the interpretation of the historical decomposition, and the nature of our contribution relative to the existing literature. In particular, we now emphasize more clearly that the paper addresses a conditional policy question—how commodity revenue realizations affect the performance of a fixed, pre-announced fiscal consolidation plan—rather than providing an unconditional explanation of Ecuador’s business cycle or an optimal policy design exercise.

Sincerely,

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